

P1 Coin tossed n times

Prob. to get head $= p \rightarrow$ cannot use sample space (only works for ~~even~~ equally likely outcome)

E = First toss is head

F_k = Exactly k heads

Aim: Find (n, k) s.t. E and F_k are indep.

$$\Omega = \{(X_1, \dots, X_n) : X_i = H \text{ or } T\}$$

$$E = \{(H, X_2, \dots, X_n) : X_i = H \text{ or } T\}$$

$$F_k = \{(X_1, \dots, X_n) : X_i = H \text{ or } T, k \text{ of } X_i \text{ are } H\}$$

Assume that the tosses are independent.

$$\therefore P(E) = p$$

$$P(F_k) = \binom{n}{k} p^k (1-p)^{n-k}$$

Now $E \cap F_k$ = The event that the first toss is head and there are $k-1$ heads among the remaining $n-1$ tosses.

$$\therefore \underbrace{P(E \cap F_k)}_{\substack{\text{first} \\ \text{toss}}} = \underbrace{p \cdot \binom{n-1}{k-1} p^{k-1} (1-p)^{n-k}}_{n-1 \text{ tosses}}$$

If $0 < p < 1$,

E and F_k are indep. iff $P(E \cap F_k) = P(E)P(F_k)$

$$\text{iff } \underline{p} \underline{\binom{n-1}{k-1}} \underline{p^{k-1}} \underline{(1-p)^{n-k}} = \underline{p} \underline{\binom{n}{k}} \underline{p^k} \underline{(1-p)^{n-k}}$$

$$\text{iff } \binom{n-1}{k-1} = p \binom{n}{k}$$

$$\text{iff } \frac{(n-1)!}{(k-1)!(n-k)!} = p \frac{n!}{(n-k)!k!}$$

$$\text{iff } k = pn$$

$\therefore$ All pairs (n, k) s.t. $\frac{k}{n} = p$.

OR

~~All pairs~~ (n, pn) where $n \in \mathbb{N}$.

If $p=0$, then $P(E \cap F_k) = 0 = P(E)P(F_k) \Rightarrow E$ and F_k ~~are~~ indep. for all n, k EN

If $p=1$, then $P(E)=1$, $P(F_k) = \begin{cases} 1 & \text{if } k=n \\ 0 & \text{if } 0 \leq k < n \end{cases} \Rightarrow E$ and F_k indep. for all n, k 7/0